# ACE1117

## 1A Positive Voltage Regulators

### Description

ACE1117 series of high performance low dropout voltage regulators are designed for applications that require efficient conversion and fast transient response.

In addition, ACE1117 is designed to be stable under conditions where Cin and Cout are not present. However, it is recommended to include Cin and Cout in the system design as this will speed up the transient response and increase the PSRR rating.

### Features

- Low Dropout Performance.
- Low Quiescent Current: 3mA (Typ.)
- Guaranteed 1A Output Current.
- Wide input Supply Voltage Range.
- Stable operation without Cin and Cout.
- Over-temperature and Over-current Protection
- Fixed or Adjustable Output Voltage.
- Rugged 2KV ESD withstand capability.
- Available in SOT-223 and TO-252 Packages.
- RoHS Compliant and 100% Lead (Pb)-Free

### Application

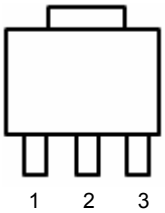
- Active SCSI Terminators
- High Efficiency Linear Regulators
- 5V to 3.3V Linear Regulators
- Motherboard Clock Supplies

### Absolute Maximum Ratings

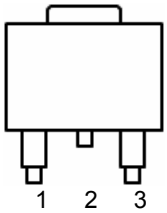
| Parameter | Symbol | Max | Unit |
|---|---|---|---|
| Input voltage | $V_{IN}$ | 9 | V |
| Junction temperature | $T_J$ | -40 to 130 | °C |
| Thermal Resistance Junction to Ambient | $\theta_{JA}$ | 60 | °C/W |
| Storage temperature | $T_S$ | - 40 to 150 | °C |

### Packaging Type

SOT-223




TO-252




| Pin | Symbol |
|---|---|
| 1 | ADJ/GND |
| 2 | Vout |
| 3 | Vin |

## Ordering information

Selection Guide

ACE1117 XX XX + H

- H: Halogen - free
- +: Pb - free
- XX: XM : SOT-223, YM : TO-252
- XX: Output Voltage : 1.5 / 1.8V / 2.5V / 3.3V….. 5.0V / Default: Adjustable Version

## Electrical Characteristics

$V_{IN.MAX} \leqq 8V$, $V_{IN.MAX}$-$V_{OUT}$=1.5V, $I_{OUT}$=10mA, $C_{IN}$=10uF, $C_{OUT}$=22uF, $T_A$=25℃, unless otherwise specified.

| Parameter | Symbol | Test Conditions | Min | Typ | Mum | Unit |
|---|---|---|---|---|---|---|
| Output Voltage (1) | $V_O$ | ACE1117-15 | 1.470 | 1.5 | 1.530 | V |
| | | ACE1117-18 | 1.764 | 1.8 | 1.836 | |
| | | ACE1117-25 | 2.450 | 2.5 | 2.550 | |
| | | ACE1117-33 | 3.234 | 3.3 | 3.366 | |
| | | ACE1117-50 | 4.900 | 5.0 | 5.100 | |
| Reference Voltage (1) (ADJ. Voltage Version) | $V_{REF}$ | ($V_{IN}$-$V_{OUT}$)=1.5V $I_{OUT}$=10mA | (-2%) | 1.250 | (+2%) | V |
| Line Regulation (1) | $V_{SR}$ | $V_{OUT}$+1.5V<$V_{IN}$<8V $I_{OUT}$=10mA | | 0.3 | | % |
| Load Regulation (1) | $V_{LR}$ | ($V_{IN}$-$V_{OUT}$)=1.5V 10mA$\leqq I_{OUT} \leqq$1A | | 0.5 | | % |
| Quiescent Current (2) | $I_Q$ | Fixed Output Version | | 3 | | mA |
| Adjust Pin Current | $I_{ADJ}$ | | | 65 | | uA |
| Adjust Pin Current Change | $\triangle I_{ADJ}$ | $V_{OUT}$+1.5V<$V_{IN}$<8V 10mA$\leqq I_{OUT} \leqq$1A | | 14 | | uA |
| Dropout Voltage (3) | $V_D$ | $I_{OUT}$=1A | | 1.3 | | V |
| Minimum Load Current | $I_O$ | | | 0.4 | | mA |
| Current Limit | $I_{CL}$ | | | 1.8 | | A |
| Temperature Coefficient | $T_C$ | | | 0.07 | | %/℃ |
| Thermal Protection | OTP | | | 175 | | ℃ |
| RMS Output Noise | $V_N$ | $T_A$=25℃, 10Hz$\leqq$f$\leqq$100kHz | | 0.003 | | %$V_O$ |
| Ripple Rejection Ratio | $R_A$ | f=120Hz, COUT=22uF (Tantalum), ($V_{IN}$-$V_{OUT}$)=3V, $I_{OUT}$=10mA | | 55 | | dB |

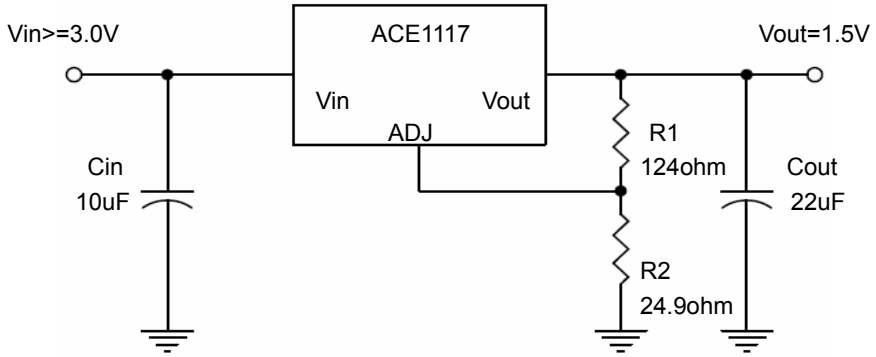
Note1: Low duty cycle pulse testing with which $T_J$ remains unchanged.

2: The $I_Q$ of ACE1117-1.5 is 1.2mA (Typ.)

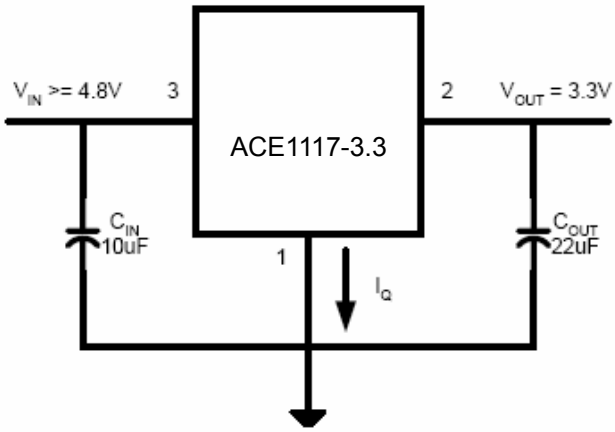
3: The dropout voltage is the input/output differential at which the circuit ceases to regulate against further reduction in input

## Typical Application

Adjustable Voltage Regulator




$V_{OUT}=V_{REF} * (1+R2/R1)+I_{ADJ} * R2$




Fixed Voltage Regulator

## Application Hints

The typical Linear regulator would require external capacitors to ensure stability. However, ACE1117 is designed in such a way that these external capacitor can be omitted if the PCB layout is tight and system noise is not very high. For better transient and PSRR performance, the Input and Output capacitors are still recommended.

## Input Capacitor

An input capacitor of 10μF is recommended. Ceramic or Tantalum can be used. The value can be increased without upper limit.

## Output Capacitor

An output capacitor of 22uF is recommended for better transient and PSRR performance. It should be placed no more than 1 cm away from the $V_{OUT}$ pin, and connected directly between $V_{OUT}$ and GND pins. The value may be increased without upper limit.

## Thermal Considerations

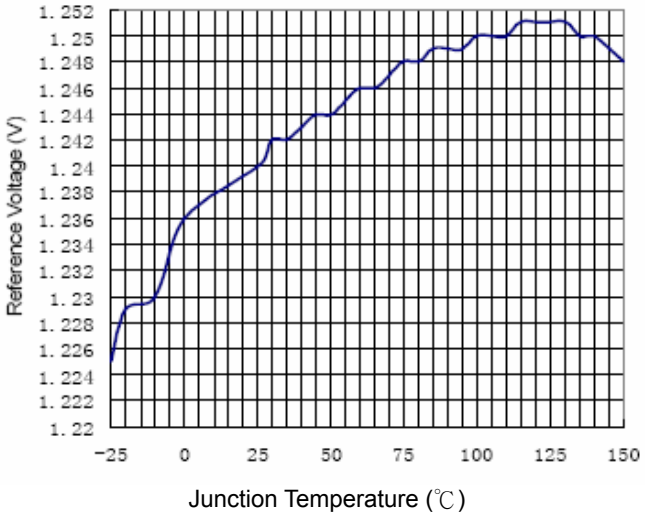
It is important that the thermal limit of the package is not exceeded. The ACE1117 has built-in thermal protection. When the thermal limit is exceeded, the IC will enter protection, and VOUT will be pulled to ground. The power dissipation for a given application can be calculated as following:

The power dissipation ($P_D$) is $P_D = I_{OUT} * [V_{IN} – V_{OUT}]$

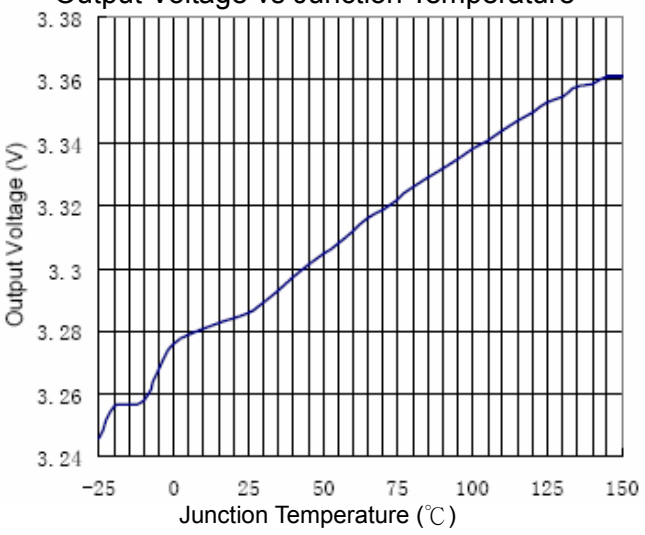
The thermal limit of the package is then limited to $P_{D(MAX)} = [T_J – T_A]/\Theta_{JA}$ where $T_J$ is the junction temperature, $T_A$ is the ambient temperature, and $\Theta_{JA}$ is around 180°C/W for ACE1117. ACE1117 is designed to enter thermal protection at 175°C. For example, if TA is 25°C then the maximum PD is limited to about 0.83W. In other words, if $I_{OUT(MAX)}$ = 500mA, then $[V_{IN} – V_{OUT}]$ can not exceed 1.66V. (Ref. SOT-223 without heat sink.)

## Typical Performance Characteristic

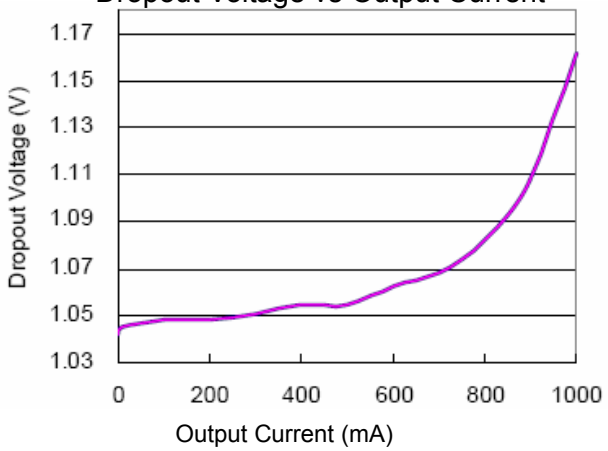
Reference Voltage vs Junction Temperature



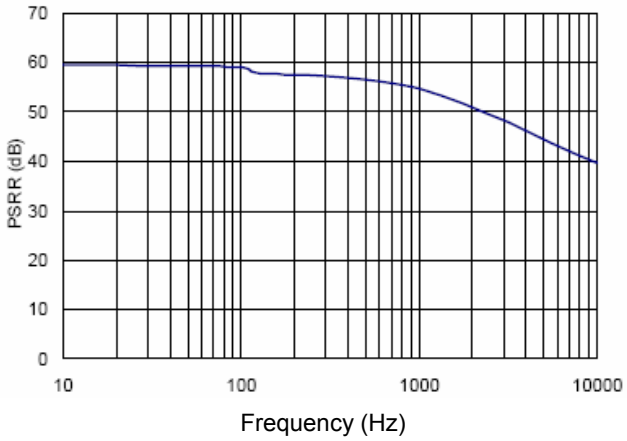

Output Voltage vs Junction Temperature



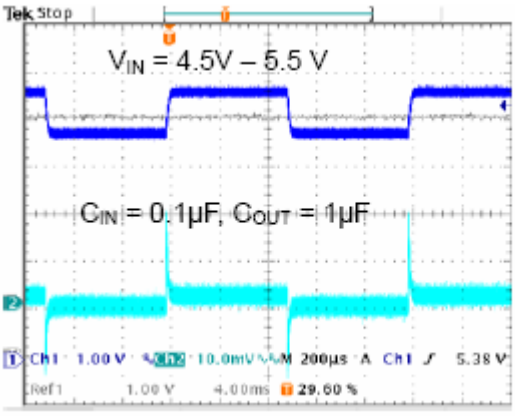
Dropout Voltage vs Output Current



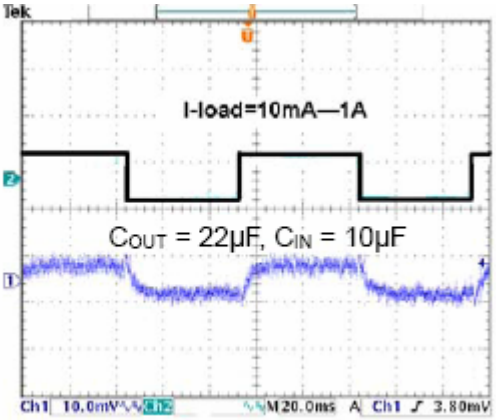

PSRR vs Frequency



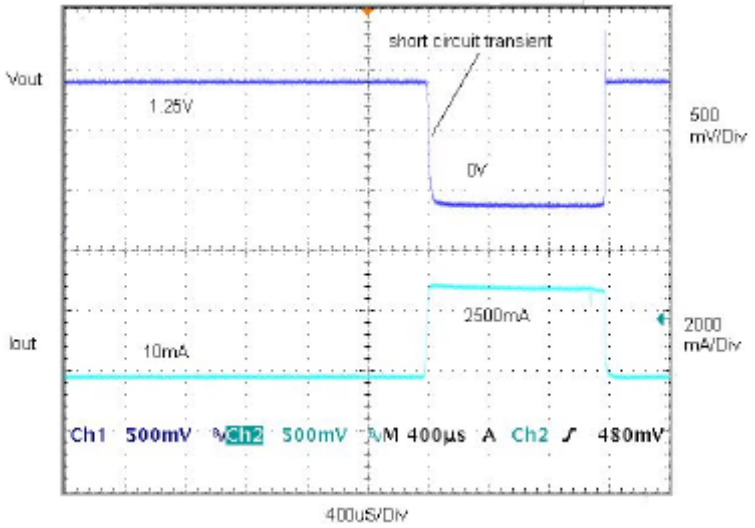

Liine Transients




Load Transients

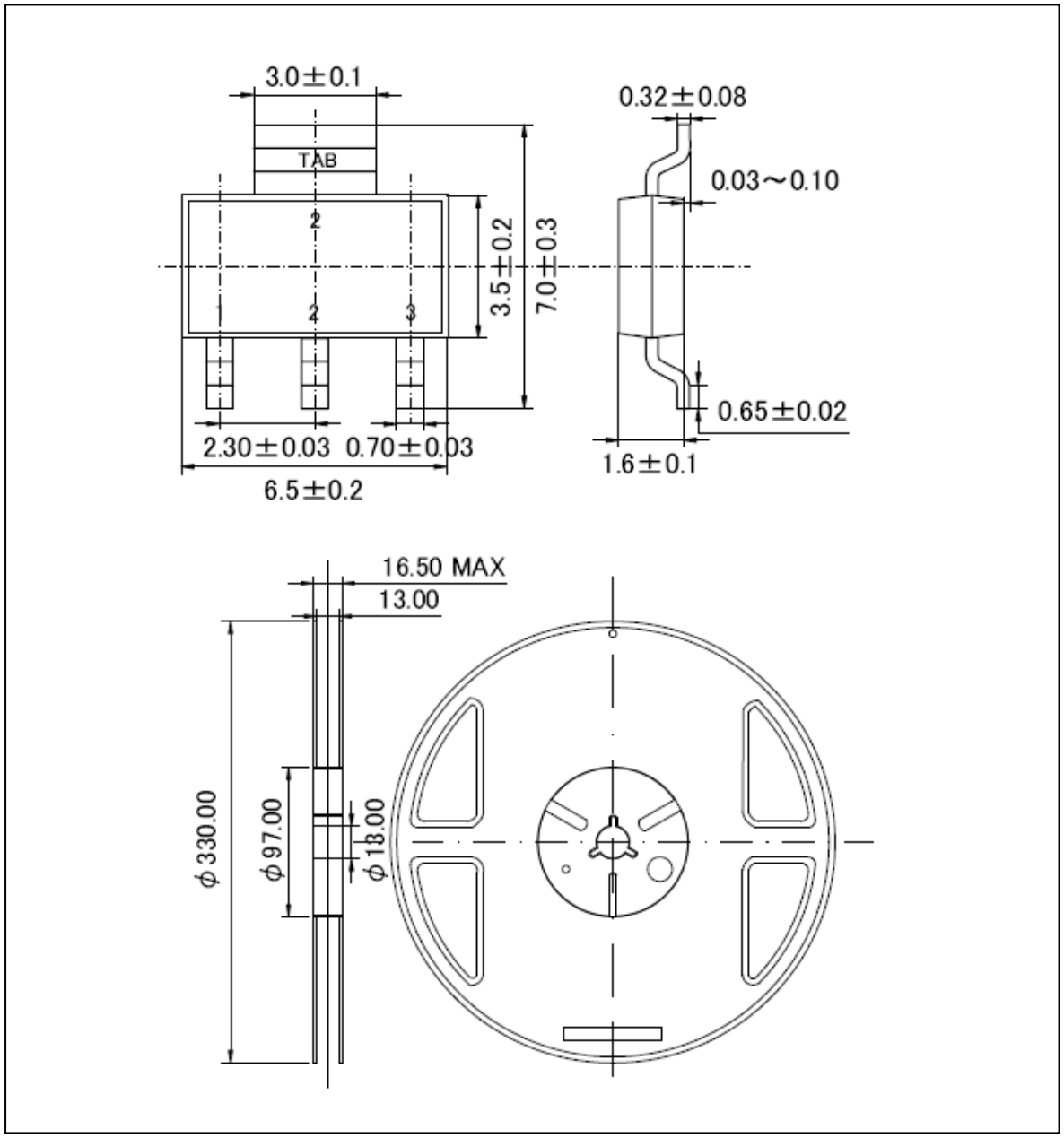



Short Transient Response



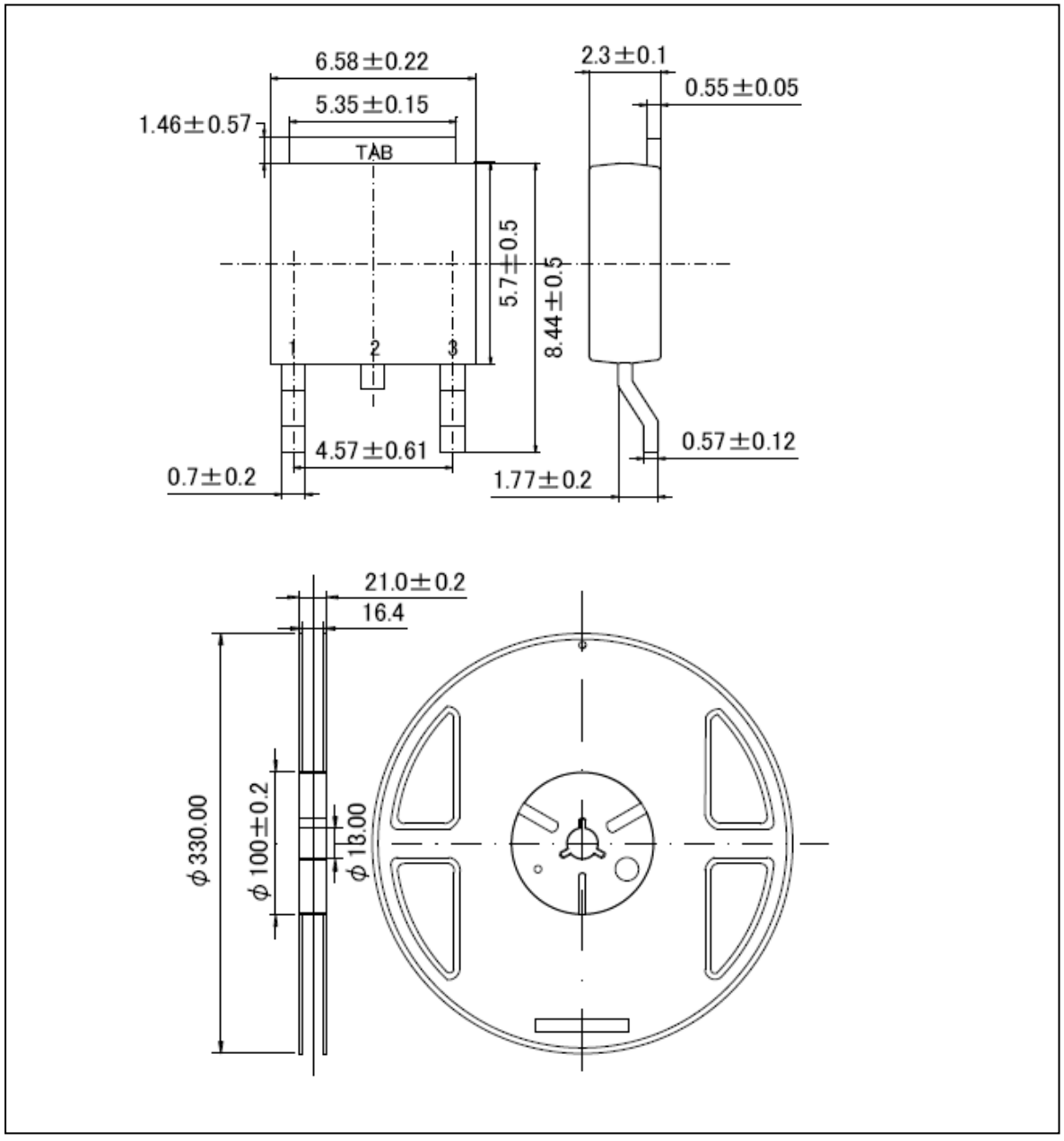
## Packing Information

### SOT-223



## Packing Information

### TO-252



Notes

ACE does not assume any responsibility for use as critical components in life support devices or systems without the express written approval of the president and general counsel of ACE Electronics Co., LTD. As sued herein:

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2. A critical component is any component of a life support device or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.

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